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Automated identification of pilot failures in aviation accidents using a BERT-based classifier and topic modeling

July B. Macedo [a,b], Plínio M.S. Ramos [a,b], Caio B.S. Maior [a,c,*], Márcio J.C. Moura [a,b], Isis D. Lins [a,b]

[a] CEERMA – Center for Risk Analysis, Reliability Engineering and Environmental Modeling, Federal University of Pernambuco, Brazil
[b] Department of Industrial Engineering, Federal University of Pernambuco, Brazil
[c] Technology Center, Universidade Federal de Pernambuco, Brazil

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ABSTRACT

Aviation accidents are a significant concern due to the potential loss of life, and human factors have emerged as the primary underlying cause. The aviation industry maintains extensive records, including accident investigation reports, which offer valuable insights for decision-making. This research explores the application of Natural Language Processing (NLP) as a solution for analyzing these documents and conducting risk assessments, empowering experts to manage accidents better and develop effective preventive measures. We propose a novel methodology that leverages a Bidirectional Encoder Representations from Transformers (BERT)-based classifier, combined with topic modeling techniques, to automate the labeling of accident datasets and identify key pilot failures contributing to aviation accidents. This automated labeling process is a critical step in efficiently creating a high-quality dataset essential for training a classifier capable of accurately detecting specific failure types. By applying the methodology to accident reports from the National Transportation Safety Board (NTSB), we successfully trained a classifier that identifies pilot failures, such as skill-based errors, routine violations, and perceptual errors. This study contributes to the field by introducing an innovative integration of contextual embeddings and topic modeling, significantly reducing manual efforts in data preparation while enhancing the precision and efficiency of analyzing aviation accident data. The findings demonstrate the potential of NLP to streamline the analysis of accident reports, assisting experts in developing targeted training programs, procedural improvements, and risk mitigation strategies to address pilot-related errors effectively.

1. Introduction

Risk Analysis (RA) is a critical concern across different industries as it enables experts to manage risks proactively and prevent potential accidents [1,2]. In the aviation industry, accidents frequently result in significant financial losses and loss of life [3]. Fig. 1 illustrates the number of aviation crashes and fatalities attributable to aviation accidents in the last ten years, as reported by the Bureau of Aircraft Accidents Archives (ACRO) [4,5]. In this context, approximately 80 % of all aviation accidents are primarily caused by human errors [6]. Indeed, despite a reduction in the number of annual accidents over the last few decades, Human Factors (HF) have long been recognized as a significant contributor to aviation incidents and accidents [7]. According to the European Union Aviation Safety Agency (EASA), Human Factors consistently play a critical role in the causes of aviation accidents, particularly in the areas of decision-making, communication, and crew resource management [8].

To address the prevalence of human error in accident causation, the Human Factors Analysis and Classification System (HFACS) has been developed as a structured framework for accident investigation. Originating from the U.S. Navy and inspired by Reason's Swiss-cheese model of accident causation, HFACS categorizes human errors into four hierarchical layers: Unsafe Acts, Preconditions for Unsafe Acts, Unsafe Supervision, and Organizational Influences [9,10]. This framework distinguishes between active failures, which are immediate precursors to accidents, and latent failures, which are systemic organizational vulnerabilities. To systematically analyze and address the root causes embedded within organizational structures, HFACS aids in identifying

* Corresponding author.
E-mail addresses: july.bias@ufpe.br (J.B. Macedo), plinio.marcio@ufpe.br (P.M.S. Ramos), caio.maior@ufpe.br (C.B.S. Maior), marcio.cmoura@ufpe.br (M.J.C. Moura), isis.lins@ufpe.br (I.D. Lins).

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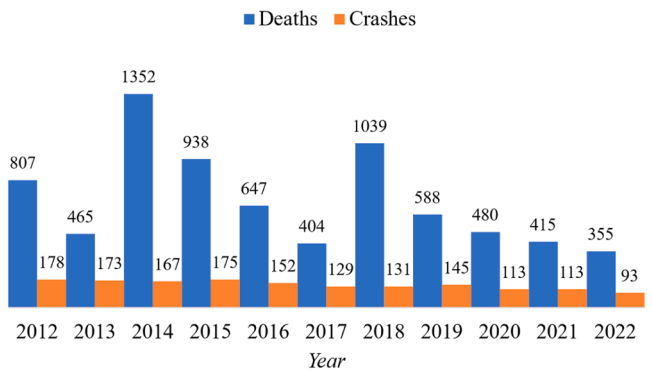


Fig. 1. Number of accidents and deaths rate per year according to ACRO (“ [4,5]).

trends, implementing targeted interventions, and improving overall safety performance. The latest HFACS 8.0 framework, proposed by the U.S. Department of Defense [11], underscores its adaptability and relevance across various sectors.

In addition, regulatory agencies mandate companies to maintain a compilation of accident reports to enable safety specialists to propose measures to mitigate or eliminate identified root causes following their analysis [1]. An accident investigation aims to uncover and report the underlying causes of a particular accident [12]. These reports are generally composed in natural language, and a comprehensive human review of the entire compilation of reports is nearly impractical given the large volume of reports produced by a company [13].

In this context, Natural Language Processing (NLP) techniques hold promise in supporting RA since they can be applied to extract, organize, and classify information from a text [14]. Data-driven approaches are increasingly employed to enhance RA. Indeed, Li et al. [15] mentioned the potential benefits of state-of-the-art NLP models and described how such models can support prognostics and health management. Recent studies have explored NLP for accident analysis across various industries [16–21]. However, existing approaches primarily use frequency-based models, such as BoW and TF-IDF, which lack contextual understanding. Additionally, many prior works require manual labeling of accident causes, which is time-intensive and subjective.

This study applies NLP techniques to detect and analyze key pilot failures contributing to aviation accidents within accident narratives. It performs topic modeling with contextual word-vector representations derived from pre-trained Bidirectional Encoder Representation from Transformers (BERT) [22] to identify pilot failures. This study overcomes the inherent limitations of traditional topic modeling approaches like Latent Dirichlet Allocation (LDA) and Structural Topic Model (STM) [23] by using these contextual embeddings and using HDBSCAN, a clustering algorithm that does not require the predefinition of the number of topics. While human factors broadly contribute to aviation accidents - including organizational, supervisory, and maintenance-related causes - this study focuses on pilot-related human errors (hereafter referred to as ‘pilot failures’) to demonstrate how NLP methods can automatically extract such events from accident narratives.

A contribution of this research is the semi-automated annotation framework enabled by the identified topics, which reduces the effort required for manual labeling of accident narratives. This annotated dataset is then used to develop a BERT-based model for classifying pilot failures in aviation accident reports. Finally, the proposed methodology is applied to a publicly available database of accident investigations conducted by the NTSB from 1982 to 2022 [24]. By combining contextual embeddings, advanced clustering techniques, and supervised learning, this study provides a novel approach to analyzing aviation accident narratives, facilitating the identification of critical pilot failures and enabling the development of targeted preventive measures.

The manuscript is organized as follows: Section 2 presents a literature review of studies on the application of NLP in the context of aviation accidents. In Section 3, key concepts of language models are introduced and described to facilitate understanding of the proposed methodology and provides detailed information about the aviation accident database and describes the proposed topic modeling approach based on contextual embeddings, as well as the development of the pilot failures classifier. Section 4 applies the topic modeling to identify pilot failures in the aviation accident database, creates a labeled dataset used to train the proposed classifier based on BERT, and presents a detailed analysis of the results obtained. Finally, Section 5 provides concluding remarks.

2. Literature review

Accident causation models such as the HFACS [25] and the Swiss Cheese Model [26] have been widely used to analyze aviation accidents. These models emphasize the role of human errors, latent conditions, and organizational influences in accident progression. However, applying these frameworks requires extensive manual classification, which limits their scalability in large datasets. By integrating NLP techniques, our study aims to automate the identification of pilot failure patterns, making it possible to extract structured insights from unstructured accident narratives.

NLP-based methods have been increasingly employed to analyze accident investigation reports across various domains. Ahmadpourgeshlagi et al. [16], Baker et al. [17], Janstrup et al. [18], Kutela et al. [19], Kwayu et al. [20], and Lu et al., [21] have utilized NLP to extract meaningful insights from these reports. Moreover, Abdullah et al. [27] and Song et al. [28] applied LDA topic modeling to analyze motor vehicle crash records and estimate train derailment severity, respectively. Although NLP has been applied to risk assessment in multiple industries, limitations persist, particularly in the context of aviation accidents.

The current literature lacks studies that use NLP techniques to investigate the specific human actions that lead to aviation accidents from accident narratives. For example, Rodrigues et al., [29] attributed pilot fatigue as the primary cause of aviation accidents and focused on fatigue modeling and analysis, and [30] aimed to enhance airlines’ awareness of potential runway overrun risks associated with pilot behavior. Sun and Tang [31] developed a communication error detection system by using a Bayesian network and extracting features from human communications. Although relevant to communication errors, the study did not explicitly analyze patterns of pilot failures leading to aviation accidents.

Moreover, Zhang et al. [32] used the NTSB texts to build supervised machine-learning models to diagnose adverse events like accidents, aircraft damage, or fatalities. Abdullah et al. [27] utilized semantic text categorization to reduce the chance of control tower operators’ errors. The authors recorded the speech of the operator and pilot and convert it into text. Next, the texts were semantically categorized into relevant categories for retrieving the semantics for each token. However, the authors did not provide any practical applications or validation results of the proposed model. In addition, V. P. Singh and Mahmoud [33] adopted a novel approach by transforming Human Machine Interaction state time-series patterns into a language translation task employing NLP techniques to forecast operator response patterns, but their study lacked a detailed exploration of its applicability to pilot failures in aviation accidents.

However, these studies did not focus on identifying or analyzing the underlying causes of these events, leaving a critical gap in understanding the human errors, specifically pilot failures, that contribute to aviation accidents. Identifying pilot failures that caused aviation accidents provides a comprehensive overview of potential hazards, facilitating the design of preventive measures to enhance safety. Therefore, there is a need to fill the gap in the application of NLP to aviation text data to identify pilot failures in aviation accidents.

Besides that, existing studies primarily rely on manual expert classifications or frequency-based NLP models, which do not fully capture the contextual relationships between failure patterns. For example, Robinson [34] examined the role of subject matter experts in interpreting processed aviation narratives. The author employed LDA to analyse a dataset from the aviation reporting database. While the author acknowledged the successful application of the LDA model in identifying meaningful clusters interpretable by experts, the study did not prioritise the exploration of pilot failures or similar factors specific to aviation accidents.

In addition, Rose et al. [35] applied LDA to NTSB aviation accident narratives to identify common topics. Similarly, Xiong et al. [36] examined the application of LDA for civil aviation incident reports, identifying potential risk factors [37] applied the STM to analyze aviation safety incident reports from the Aviation Safety Reporting System to identify latent topics and trends within these reports to uncover underlying safety issues. While effective in identifying general patterns, LDA and STM have inherent limitations, such as their reliance on the Bag-of-Words (BoW) approach, which disregards the contextual relationships between words within a sentence and the need to predefine the number of topics to be identified [38].

Moreover, Miyamoto et al. [39] presented a methodology that leverages aviation text data to identify the underlying causes of flight delays and cancellations, to infer operational inefficiencies. While the study focused on operational inefficiencies, including maintenance issues as a primary cause, it did not explore pilot failures and relied on traditional techniques such as BoW and TF-IDF rather than more advanced models like deep learning or transformers.

Madeira et al. [40] developed a methodology to identify and categorize factors associated with aviation incidents using descriptive text data. In this study, manual labeling of accidents was conducted, and traditional language models were utilized. Local tests were conducted, concluding that Doc2Vec performed best. Moreover, the study did not isolate pilot failures as a specific category. Perboli et al. [41] presented an approach to analyze accident investigation reports using NLP techniques, claiming to employ state-of-the-art models. However, their methodology relied primarily on traditional NLP methods like Word2-Vec and Doc2Vec, raising concerns about the effectiveness of these models in capturing complex relationships, such as those underlying pilot failures, without additional augmented features [12,42].

Our research extends these approaches by integrating BERT embeddings and unsupervised topic modeling. This approach offers better semantic understanding and avoids the need for predefined topic numbers, providing a scalable, automated method for identifying pilot failures and supporting data-driven accident prevention strategies. Also, the existing literature reveals a scarcity of studies that apply NLP techniques to identify pilot failures as distinct from broader human factors in aviation accidents. Addressing this gap is crucial, as pilot failures represent a significant contributor to aviation incidents and could benefit from targeted analysis using advanced NLP methodologies.

3. Methodology

The proposed methodology can be summarized in three main parts, as depicted in Fig. 2. Firstly, a data analysis step was conducted, involving the examination of the aviation database to determine the study's scope and selection of relevant information. Subsequently, a topic modeling process was performed to cluster similar accident causes, extract the main topic for each cluster, and identify the associated pilot failure for each accident group. Lastly, a pilot failure classifier was developed by fine-tuning a pre-trained BERT model using accident narratives that were labeled with the pilot failures identified in the previous step. All text data extracted from the database was in English, and the entire methodology was implemented using Python.

3.1. Aviation accidents data analysis

The NTSB investigates and reports unintentional transportation incidents, including aviation accidents, specific road accidents, maritime accidents, and incidents involving the release of hazardous materials during transportation. The NTSB's aviation accident database comprises data from completed investigations of civil aviation accidents and incidents in the United States. Generally, an initial report is swiftly made available after an accident, later supplanted by a final account of the incident along with its probable cause. This database is open to the public, and our analysis is centered on aviation accident investigation reports conducted from 1982 to October 2022 [24].

The database contains over 20,000 documents organized into spreadsheets, providing comprehensive information on accidents, including the date of occurrence, the number of people injured, the severity of injuries, and whether the individuals affected were crew or passengers. Our focus lies on the "accident description", a more detailed narrative typically containing information on the accident's cause and its impacts. Additionally, within the database, one can find a succinct narrative detailing the accident's cause (referred to in this manuscript as the "accident cause description"). However, this shorter description of causes is not always available and is not standardized, which makes it difficult for experts to extract insights about common causes of accidents and propose effective preventive measures.

Fig. 3 visually represents how data is used across different stages of our methodology. This figure illustrates how accident descriptions undergo preprocessing, topic modeling for failure identification, and annotation before being split into training, validation, and test datasets for classifier development.

The topic modelling based on BERT embeddings was first applied to the accident cause descriptions to extract semantically coherent topics representing distinct types of pilot-related failures. Each resulting topic label was then transferred to the corresponding accident description of the same event and used as the target variable for fine-tuning the BERT classifier. This sequential design allows the unsupervised topic modelling step to provide interpretable labels for training the supervised BERT model, effectively bridging the topic modeling and classification stages.

Initially, preprocessing operations were performed to ensure that accident descriptions were clean, standardized, and suitable for analysis. We implemented a systematic pipeline that involved key operations such as: first, all text was converted to lowercase to eliminate inconsistencies caused by case sensitivity, ensuring uniformity across the dataset. Then, we removed numeric characters and punctuation, as they were deemed irrelevant for the semantic analysis of the accident narratives.

Next, the text was tokenized into individual words using the Natural Language Toolkit (NLTK). Tokenization enabled precise manipulation of the text, allowing for further cleaning and filtering. To refine the dataset further, stopwords (i.e., common words that typically carry little semantic meaning, such as "the," "and," and "is") were removed. This step ensured that the focus remained on terms directly related to accident

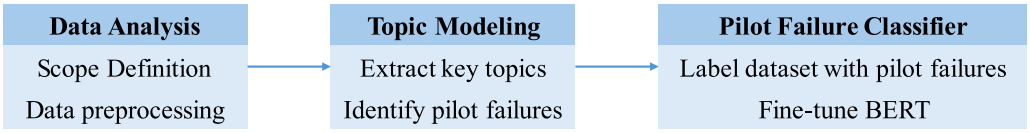


Fig. 2. Overview of the proposed methodology.

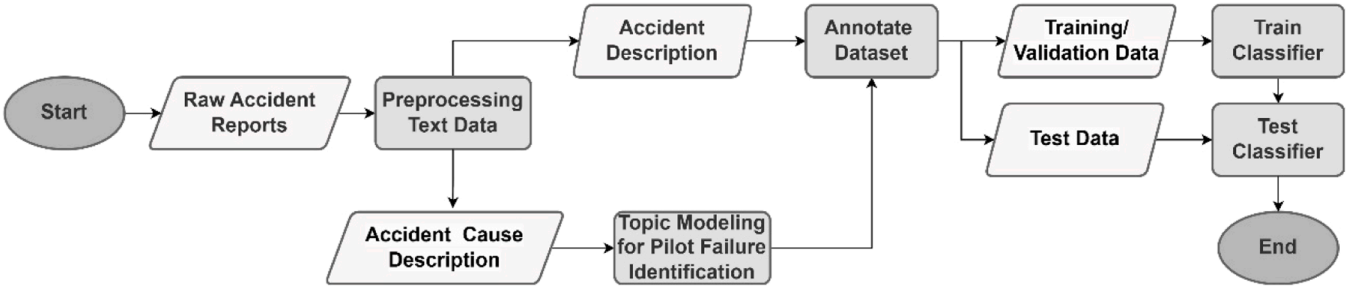


Fig. 3. Overview of how accident descriptions are processed, labelled, and used in different stages of the methodology.

causes, thereby reducing noise in the data.

After preprocessing, the database was filtered to retain only records containing both an "accident description" and a "cause description". To align the dataset with the study's objective of identifying and analyzing pilot failures, we performed an automated filtering to select accidents explicitly associated with pilot failure. Specifically, we searched for the phrase "pilot's failure" within the descriptions, isolating accidents where pilot actions played a critical role. This resulted in a final dataset of 10,530 accidents, ensuring that the analysis remained focused on cases relevant to the study's goals.

The preprocessing pipeline was implemented in Python, using libraries such as Pandas and NLTK for efficient manipulation and cleaning of data. This systematic approach ensured that the textual data were free of irrelevant elements and prepared for subsequent analysis, such as topic modeling and classification.

3.2. Topic modeling

Developing an automated classifier for accident causes, directly from the "accident description", would benefit experts. It would speed up the evaluation process, revealing patterns and common causes efficiently. However, training a supervised classifier requires a labeled accident database, which can be challenging, especially when causes are presented as descriptions without clear categorization [43]. Thus, we employed topic modeling on the "accident cause descriptions" to identify common causes and subsequently labeled the database with reduced effort. It is worth noting that the dataset was pre-filtered to include only accidents attributable to pilot failures. Accordingly, the topic modeling step aimed to identify subcategories of pilot error rather than distinguishing pilot-related accidents from other causes.

Topic modeling techniques provide valuable insights to identify common themes and the underlying narrative within a collection of documents. Traditional models like LDA and its derivatives [44] represent documents as BoW and model them as mixtures of latent topics. However, BoW representations cannot capture word context within sentences. In contrast, BERT [22] has demonstrated impressive results in generating contextual word and sentence embeddings [45]. To overcome the limitations of LDA and other frequency-based models, we applied a BERT-based topic modeling approach combined with HDBSCAN clustering as depicted in Fig. 4 [46]. Unlike LDA, which

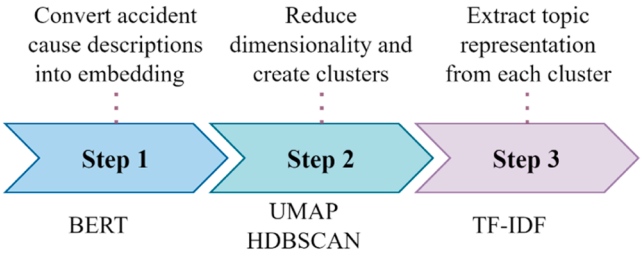


Fig. 4. Topic modeling approach based on Grootendorst [46].

requires a pre-defined number of topics, our approach automatically determines topic groupings based on the structure of the data.

To begin, we transform each "accident cause description" into an embedding representation using a pre-trained language model for comprehensive document-level information. Next, we employ UMAP [47] to reduce the dimensionality of the embeddings, optimizing clustering. Subsequently, HDBSCAN [48] is utilized to generate semantically similar clusters of accident causes, each representing a distinct topic. Finally, we extract topic representations from document clusters using a modified TF-IDF [49], allowing us to discern and analyze the underlying topics within the accident cause descriptions.

To analyze "accidents cause descriptions" and capture their semantic similarities, we utilize vector space embeddings. In step 1, we employ the pre-trained model, specifically the 'BertForSequenceClassification' implementation from the transformers library [50]. By inputting the "accident cause descriptions" into BERT, we extract 768-dimensional feature vectors for each word in the description from the final layer, as depicted in Fig. 5.

To enable clustering, it is necessary to represent each document with a single vector. To achieve this, we compute the mean of the embedding vectors associated with the words in the document, resulting in a single vector representation for each document.

Subsequently, we applied dimensionality reduction using UMAP to obtain clusters. Although t-SNE [51] and PCA [52] are commonly used algorithms for dimensionality reduction, UMAP was chosen for its ability to preserve global structure in a shorter computational time when projecting to lower dimensions [53].

In step 2, we utilized HDBSCAN to perform the clustering task, which is an unsupervised technique that can identify related groups within a dataset. One of the advantages of HDBSCAN is that it does not require the number of clusters to be specified as input, making it suitable for datasets where the number of clusters is unknown. Additionally, HDBSCAN is capable of identifying dense clusters, allowing for the possibility of unassigned data points that are considered outliers [48, 54]. For our implementation, we utilized the UMAP and HDBSCAN implementations provided by the scikit-learn library [55].

Next, in step 3, we applied a class-based variation of TF-IDF to the clusters obtained in step 2. This variation involves applying the TF-IDF formula (Eq. (2)) to multiple classes by combining multiple documents per class. In our case, each cluster represents a class, and we computed the TF-IDF values accordingly.

$$W_{w,c} = |tf_{w,c}| \times \log\left(1 + \frac{A}{f_w}\right) \tag{1}$$

where $tf_{w,c}$ is term frequency, which is the frequency of each word w is extracted for each class c and normalized. The variable A represents the total number of documents, while f_w represents the document frequency of word w across all classes. In this process, each cluster is transformed into a single document instead of a collection of multiple documents.

Consequently, a single importance value $W_{w,c}$ is assigned to each word w in a cluster c, which enables the creation of a topic. By selecting the top n most important words in each cluster, we obtain a reliable

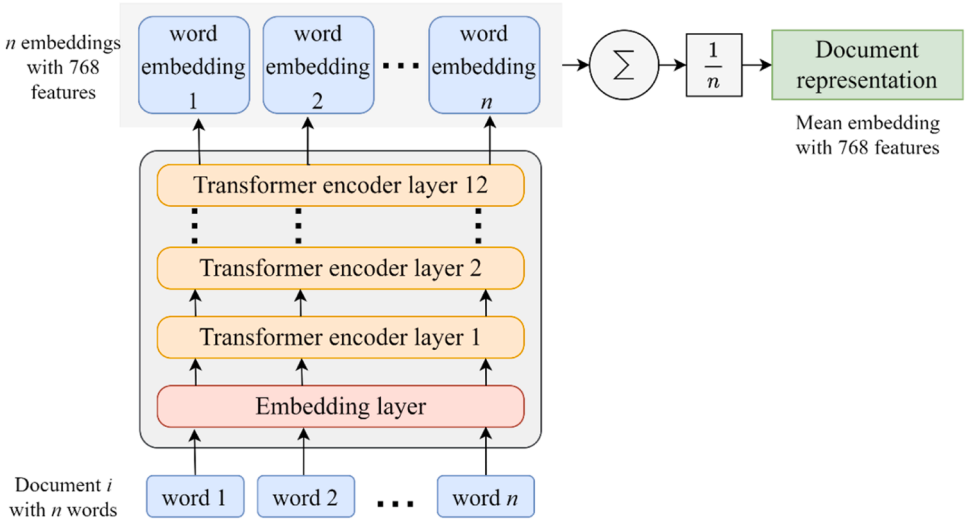


Fig. 5. Definition of documents' representation based on word embeddings extracted from pre-trained BERT.

representation of the cluster and its corresponding topic. In this study, we chose to extract the top 5 words to capture the essence of each cluster's main topic. Through this process, we examined the topics to identify the pilot failure associated with accidents and subsequently labeled the accidents based on the identified pilot failures.

To ensure the reliability of the topic modeling process before classification, we evaluated the topic themes using quantitative Topic Diversity, which measures the range of unique words across all topics, and Topic Entropy, which reflects the distribution of topic probabilities across the dataset. These metrics validate the quality of the generated topics and confirm their suitability for downstream classification tasks.

Subsequently, we utilized the annotated dataset to fine-tune a pre-trained BERT model for a supervised task: the identification of pilot failures based on an "accident description", which includes information about environmental conditions, consequences, and other event-related details. It is crucial to highlight that the topic modeling was conducted using the "accident cause descriptions", while the model was fed with the "accident description" as input. Although a single dominant topic is assigned to each report for classifier training, we retain the complete posterior topic-probability vector for downstream analyses. This allows secondary topics to be examined and supports extensions to multi-label or hierarchical classification.

To validate the semantic coherence and robustness of the automatically generated labels, we compared the resulting topics with classical bag-of-words topic models (STM and LDA) using the same preprocessing and vocabulary. This comparison aims to verify that the major thematic structures are representation-invariant and not artifacts of the embedding model.

Although topic diversity and entropy were reported for diagnostic purposes, they were not used as standalone optimization criteria. Parameter selection was guided by a multi-step ablation process combining HDBSCAN sensitivity analysis, coherence metrics, cross-model validation (LDA/STM), semantic topic alignment, and downstream classifier performance. The following section outlines the implementation and training process of the classifier.

3.3. Modeling process

Transformers-based models have demonstrated their efficacy in NLP by acquiring universal language representations through training on extensive text corpora. However, training transformers from scratch entails substantial computational expenses and time requirements [56]. BERT and Generative Pre-trained Transformer (GPT) were pioneering pre-trained language models that utilized transformer encoders and decoders, respectively [22].

In the context of risk and reliability engineering, despite the advances when considering studies related to core NLP tasks, it is common to find applications using classical models such as BoW, TF-IDF, and Doc2Vec. Moreover, BERT implementations in Pytorch and Tensorflow have been available for more than a year [50], in different languages, stably, with no long-term compatibility problems between libraries. Hence, we developed our methodology based on BERT, as it gives us flexibility and robustness.

Multiple variations of pre-trained BERT models are accessible for download, allowing users to fine-tune these models for specific supervised learning tasks [57]. This involves adding an untrained layer of neurons on top of the pre-trained BERT model. Overall, during fine-tuning, the pre-trained parameters are adopted to initialize the model, which is then updated using labeled data specifically tailored to the supervised task at hand [58]. For instance, we can adjust BERT's architecture by adding one output layer on top of the pre-trained model to adapt it for performing a classification task. For instance, we can adjust BERT's architecture by adding one output layer on top of the pre-trained model to adapt it for performing a classification task. It's worth mentioning that the parameters related to the additional layer are the only parameters that require random initialization and learning from scratch. This approach enables the construction of state-of-the-art architectures within a reasonable timeframe [59]. For more details see [22].

Thus, the human factor classifier was built fine-tuning the Pytorch implementation of pre-trained BERT available at transformers library [50], which provides APIs to quickly download and use several pre-trained models on a given text, and thus fine-tune them on our own datasets. We here adopted the 'bert-base-multilingual-cased' version of the 'BertForSequenceClassification' model that is a version pre-trained in 104 languages.

To adapt BERT for our classification task, we added an output layer on top of the pre-trained model. This layer uses the final hidden state $\boldsymbol{h}$ of the [CLS] token (a special classification token) to represent the input sentence and compute probabilities using a *softmax* activation function (2).

$$p(c|\boldsymbol{h}) = activation\left(\boldsymbol{h}^T\boldsymbol{W} + b\right) \tag{2}$$

where c is the class of the input sentence, b is the bias, and $\boldsymbol{W}$ is the weights matrix of the added output layer. Moreover, we added because the model classifies the input sentence into a single class among mutually exclusive classes. Then, we used a *softmax* to predict the probability

of each label c. Simply put, the *softmax* activation function outputs are mutually exclusive, and then the sum of their probabilities is 1; thus, the predicted class is the one, which has the highest probability [60]. The categorical cross-entropy loss function was employed (see [61] for details), and the BertAdam optimizer from the Transformers library was used for optimization.

Finally, we fine-tuned the pre-trained model, using the labeled dataset for the classification task. The dataset were randomly divided into training (80 %) and test (20 %) data, with a further 20 % of the training data set aside for validation. In this case, the loss is propagated through the entire architecture and all BERT pretrained parameters as well as $\boldsymbol{W}$ and b are fine-tuned and updated based on the new dataset. However, $\boldsymbol{W}$ and b are the only parameters that need to be randomly initialized and learned from scratch.

All experiments were conducted on a GeForce RTX 2080 Ti GPU, with approximately 4–5 h required for fine-tuning the model. We adopted a learning rate of 10^{-6}, batch size of 16, and warm-up of 0.1 to train the models for 50 epochs. Additionally, we computed precision, recall, and F_1-score. The results achieved with each model are discussed in the following section.

It is important to note that our methodology is designed as a decision-support tool rather than a replacement for expert judgment. Accident causality is inherently complex, often involving multiple contributing factors. While our model identifies a primary pilot failure per incident, this does not imply that other contributing factors are irrelevant. Instead, the model provides structured insights that can help investigators in recognizing common failure patterns, supporting targeted training and safety interventions. Future extensions of this work could incorporate a multi-label classification model to better capture interdependencies between causes.

4. Results

Firstly, to ensure effective embedding representation, we analyzed the distribution of accident cause description lengths within the dataset. Most records (75 %) contain fewer than 200 tokens (Fig. 6), making them well-suited for processing within BERT's 512-token limit. For longer descriptions, truncation was applied where necessary, prioritizing key information typically found at the beginning of accident reports.

The preprocessed "accident cause descriptions" related to pilot failures were converted to embeddings. The dimensionality of the embeddings was reduced from 768 features to 5 features. The reduced embeddings were used to train the HDBSCAN algorithm, resulting in 98 clusters. Due to the high number of clusters, we employed cosine similarity to merge similar clusters and facilitate the subsequent cluster analysis [54].

Thus, 7 clusters were obtained, including the cluster '–1′, with 5913 instances, containing instances that were not assigned to any specific cluster. To address the potential issue of high rates of outliers inherent in topic modeling methods, we employed the BERTopic library and applied *.update_topics* function from BERTopic library. This adjustment redistributed outliers in order to improve coherence across topics.

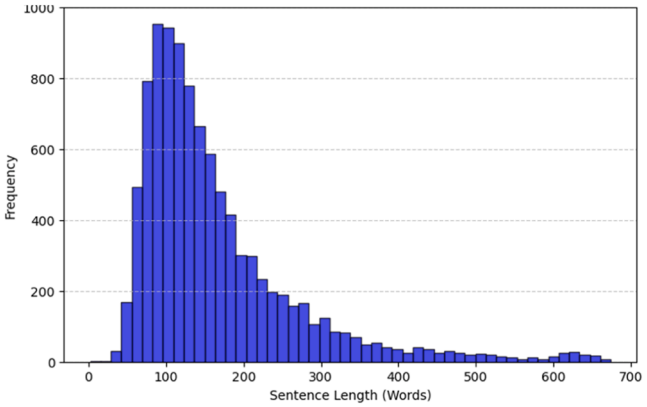


Fig. 6. Distribution of accident description lengths as number of tokens.

Table 1 presents the updated distribution of data across topics. This refinement underscores the reliability of the BERTopic approach in analyzing aviation accident data while maintaining a clear thematic structure. Following the integration of the *.update_topics* function in BERTopic, the number of data points assigned to each topic has been significantly optimized.

To evaluate the robustness of the clustering process, we performed a sensitivity analysis varying the HDBSCAN parameters: min_cluster_size = [50, 100, 150, 200] and min_samples = [5, 10, 20]. As shown in Table 2, the proportion of unassigned documents (topic −1) remained stable across configurations (≈10–30 %), confirming that this behavior results from the high redundancy of accident narratives rather than from model instability.

Parameter combinations that minimized unassigned cases tended to increase the number of clusters (up to 25), many of which were semantically redundant. More conservative settings yielded fewer, broader, and more coherent clusters. The adopted configuration (min_cluster_size = 150, min_samples = 5) provided an interpretable balance, generating 7 topics while maintaining coherence. Hence, the proportion of outliers reflects a conservative modeling choice aimed at preserving semantic clarity rather than a loss of information.

4.1. Topic analysis

The TF-IDF technique was then applied to these clusters, extracting the top 5 most significant words from each cluster based on their probabilities, which quantify the relevance of each word to the topic. These probabilities provide a clear measure of the semantic contribution of each word.

Table 3 presents the top 5 words for each topic and their respective probabilities. This approach highlights the most relevant terms while avoiding the inclusion of words with low probabilities, which may add noise or reduce interpretability. Then, our focus lies in evaluating the topics within clusters 0 to 5 to pinpoint the pilot failures responsible for accidents to establish a labeled dataset.

It is important to note that the topic modelling was applied at the document level. Each 'accident cause description' was considered as a single textual unit and assigned to one topic cluster according to the highest cluster probability. In cases where an accident description contained multiple causes (e.g., fuel-related and landing-related issues), the topic exhibiting the highest semantic similarity was selected as the dominant one. This approach ensures a one-to-one mapping between accident reports and topic clusters, allowing the resulting labels to be consistently used for training the supervised BERT classifier in Section 4.2.

Cluster 0 exhibits "failure", "maintain", and "control" as key terms, suggesting that accidents in this category are attributed to the loss of directional control. The presence of the word "directional" further reinforces this interpretation, emphasizing the challenges that pilots face in maintaining stable flight paths, particularly during takeoff and

Table 1
The number of instances in each cluster.

Cluster	Count
0	6079
1	1849
2	802
3	522
4	303
5	256

Table 2
Sensitivity analysis of HDBSCAN parameters (min_cluster_size, min_samples). Columns report total documents, number and percentage of outliers (topic −1and number of topics found (excluding −1).

min_cluster_size	min_samples	n_docs	n_neg1	% topic −1	n_topics_found
50	5	9811	3025	30.83	26
50	10	9811	1308	13.33	20
50	20	9811	927	9.45	19
100	5	9811	2054	20.94	15
100	10	9811	1461	14.89	15
100	20	9811	1184	12.07	15
150	5	9811	337	3.43	7
150	10	9811	1576	16.06	13
150	20	9811	1293	13.18	13
200	5	9811	843	8.59	4
200	10	9811	1913	19.5	11
200	20	9811	1630	16.61	11

landing.

Cluster 1 focuses on accidents related to improper landing techniques. High-probability words like “landing”, “flare”, and “improper” suggest improper execution of landing procedures, leading to incidents such as hard landings. These occurrences are often associated with insufficient training or failure to adapt to landing conditions.

Cluster 2 presents words such as “fuel”, “engine”, and “power” highlighting the central theme of fuel mismanagement. Accidents in this category often arise from issues like fuel exhaustion, contamination, or incorrect fuel selection, leading to engine failure or power loss during critical flight phases.

Cluster 3 is specific to helicopters, where terms like “helicopter”, “rotor”, and “tail” point to failures in managing critical components for flight control. Accidents in this category emphasize the unique challenges of helicopter operation, such as maintaining balance and stability. The top words in Cluster 4, such as “taxiing”, “failure”, and “lookout” indicate that accidents in this category are related to ground operations. Issues such as inadequate situational awareness or poor adherence to taxiing protocols often lead to collisions while maneuvering aircraft on the ground.

Cluster 5 top words “gear”, “landing”, and “extend” might initially seem to be purely technical; however, this cluster could also involve a human error in certain contexts. For example, the pilot could forget to extend the landing gear during the approach, possibly due to a high workload or poor checklist discipline or choose to proceed with a landing despite known mechanical warnings or abnormal indications related to the gear. Based on refined topic modeling and analysis, we identified six primary types of pilot failures contributing to aviation accidents (Table 4). These failures were further contextualized using the Human Factors Analysis and Classification System (HFACS), a structured framework that categorizes human errors into hierarchical layers.

In addition, Table 5 summarizes the coherence scores (c_v) computed for the topic sets produced by each method. One can see that STM and LDA exhibits a lower coherence (0.44 and 0.45 respectively), indicating that the primary semantic structures are consistently captured across methods.

The evaluation of the topics resulted in a Topic Diversity of 0.9083, which indicates a high level of diversity, signifying that topics are distinct and well-separated, and Topic Entropy of 2.296, which suggests that the topics are balanced and differentiated, ensuring that no single topic dominates disproportionately. This step ensures that the extracted topics provide a reliable foundation for the BERT-based classification, minimizing potential biases introduced by poorly defined topic themes.

HFACS provides a structured framework for categorizing pilot failures into Unsafe Acts, Preconditions for Unsafe Acts, Unsafe Supervision, and Organizational Influences. Our analysis places all identified failures under the Unsafe Acts category, specifically within Skill-based Errors, Routine Violations, and Perceptual Errors. Integrating topic modeling with HFACS offers a more granular understanding of these failures, revealing distinct pilot error patterns that can inform targeted interventions.

We can propose precise, category-specific interventions to reduce pilot failures and enhance aviation safety by aligning topic modeling insights with HFACS classifications:

- Training and Skill Development – Addressing Skill-based Errors (e.g., Loss of Directional Control, Hard Landings, Helicopter Directional Control) through precision training, simulator-based exercises, and high-risk phase control retention techniques.
- Procedural Reinforcement – Reducing Routine Violations (e.g., Fuel-Related Issues) by enhancing fuel management protocols, enforcing adherence to operational checklists, and implementing stricter regulatory oversight.
- Situational Awareness and Decision-Making Improvements – Mitigating Perceptual Errors (e.g., Taxiing Collisions) through enhanced

Table 4
Identified Pilot Failures and HFACS Categorization.

Pilot Failure	Description	HFACS Layer	Human Factor Category
Loss of Directional Control	Failure to maintain control during takeoff/landing	Unsafe Acts	Skill-based Errors
Hard Landings	Improper landing maneuvers with excessive impact	Unsafe Acts	Skill-based Errors
Fuel-Related Issues	Poor fuel management, exhaustion, or contamination	Unsafe Acts	Routine Violations
Loss of Helicopter Directional Control	Helicopter-specific rotor/tail control issues	Unsafe Acts	Skill-based Errors
Taxiing Collisions	Accidents during ground operations due to poor awareness	Unsafe Acts	Perceptual Errors
Landing Gear Failures	Errors in extending/inspecting landing gear	Unsafe Acts	Skill-based Errors

Table 5
Model comparison: coherence for topic models.

Model	Coherence (c_v)
BERT	0.505
STM	0.444
LDA	0.454

Table 3
The top 5 words of each cluster were obtained using TF-IDF.

Cluster	Word 1	Prob 1	Word 2	Prob 2	Word 3	Prob 3	Word 4	Prob 4	Word 5	Prob 5
0	failure	8.7	maintain	8.4	control	8.0	pilots	6.2	directional	5.8
1	landing	17.3	flare	13.9	improper	11.9	resulted	10.8	hard	7.9
2	fuel	20	engine	16.8	power	14.2	loss	11.3	due	8.2
3	helicopter	15.2	rotor	13.9	tail	7.9	control	7.8	failure	7.3
4	taxiing	19.9	taxi	17.2	failure	12.3	maintain	12.3	lookout	11.1
5	gear	34.4	landing	24.4	extend	19.1	failure	12.4	prior	8.4

situational awareness training, stricter ground operation protocols, and the use of automated taxiing aids.
- Technical and Maintenance Enhancements – Preventing mechanical failures linked to human error (e.g., Landing Gear Failures) by strengthening pre-flight inspection routines, improving maintenance schedules, and introducing automated alerts for system anomalies.

While HFACS provides a robust taxonomy for incident classification, manual HFACS coding is resource-intensive and episodic. Our automated pipeline operationalizes HFACS-compatible labels at scale, enabling near-real-time monitoring of trends. For policymakers, this means (i) evidence-based prioritization of interventions based on statistically significant trends and risk ratios, (ii) rapid detection of emergent problem subthemes that may warrant immediate investigation, and (iii) the ability to cross-reference human-factor signals with operational metadata to localize interventions (e.g., operator-specific or phase-specific training). The proposed NLP framework does not itself prescribe interventions but provides data-driven evidence of recurrent failure mechanisms that can guide domain experts in prioritizing targeted safety actions.

Beyond classification, our analysis uncovers patterns in pilot failures that may not be immediately apparent in manual assessments. For instance, our analysis identified fuel-related issues, which are often linked to mechanical failures, and revealed accidents that were caused due to procedural violations, highlighting the need for reinforced fuel management protocols. These findings illustrate how our approach can reveal actionable insights that inform targeted interventions in pilot training and operational procedures.

Although the present study presents a scalable method to convert unstructured accident narratives into HFACS-compatible indicators, assigning a single dominant label per incident is a simplification: while useful for constructing a consistent training set, multi-causal incidents exist. This approach bridges the gap between broad frameworks and practical solutions, significantly enhancing risk mitigation and improving aviation safety outcomes.

While Human Factors classification systems provide a structured approach to analyzing aviation incidents, NLP offers an additional layer of objective and consistent analysis. NLP tools can reveal trends and patterns not immediately visible through manual analysis by leveraging historical data, enhancing cross-incident analysis and supporting data-driven decision-making. Although our study demonstrates this approach using aviation accident reports, the methodology is adaptable to other domains where large volumes of unstructured data require analysis. Aviation reports serve as an ideal case study due to their standardized and publicly accessible nature, allowing for robust testing and validation of NLP methods.

4.2. Pilot failure classification

After fine-tuning BERT for 100 epochs using the labeled dataset, we evaluated the model's performance on the remaining data. reflecting robust performance in identifying pilot failures in aviation accidents. The classifier achieved an accuracy of 77.88 %, Table 6 presents the performance metrics for each pilot failure category, showing promising results in several categories despite challenges in differentiating overlapping descriptions. The confusion matrix (Fig. 7) shows the models' predictions on the test data.

Table 6
The performance on test data metric for each category.

Label	Pilot Failure	Precision (%)	Recall	F_1 score
1	Loss of Directional Control	91	79	84
2	Hard Landings	60	81	69
3	Fuel-Related Issues	64	86	73
4	Loss of Helicopter Directional Control	51	47	49
5	Taxiing Collisions	86	96	91
6	Landing Gear Failures	37	41	39

The classifier demonstrated strong performance for specific categories, such as Taxiing Collisions (91 %) and Loss of Directional Control (84 %), which are characterized by well-defined and distinct patterns in accident descriptions. Similarly, Fuel-Related Issues (73 %) achieved a robust F1 score, reflecting the model's ability to recognize descriptions involving terms like "fuel", "engine", and "loss".

However, the classifier struggled with categories such as Loss of Helicopter Directional Control (49 %) and Landing Gear Failures (39 %), likely due to the limited number of labeled examples and the inherent overlap in textual descriptions of human errors. The confusion matrix illustrates these challenges, with misclassifications between Loss of Directional Control (0) and Hard Landings (1) as well as between Helicopter Control Failure (3) and other categories. These errors highlight the need for additional contextual data or refined preprocessing methods to improve differentiation. The macro-average F1-score of 67 % and weighted average of 78 % indicate satisfactory overall performance, especially given the nuanced and overlapping nature of the categories.

We provide some examples in Table 7 to further illustrate this examples of instances misclassified by the trained model.

The first instance describes an accident where the airplane's landing gear was lowered five miles before reaching the airport during an instrument approach. However, during the short final, doubts about the gear configuration led to a belly landing, causing significant damage to the airplane. This was labeled as a *Landing Gear Failure (Cluster 5)* under Skill-based Errors, but the model misclassified it as *Loss of Helicopter Directional Control (Cluster 3)*. This misclassification likely stems from the overlap in descriptions of operational failures during the landing process, as both categories involve skill-based challenges. The second instance involves an accident during the landing flare, where the airplane experienced wind shear or a dust devil, lifting it off the runway and causing a stall. The airplane subsequently impacted the runway nose-down, causing substantial damage. Although this incident was labeled as *Loss of Directional Control (Cluster 0)* under Skill-based Errors, the model misclassified it as *Helicopter Control Failure (Cluster 3)*. The confusion likely arises from the similar nature of skill-based challenges during critical flight phases, such as maintaining control versus recovery efforts.

The third instance describes an accident involving a tailwheel-equipped airplane during a full-stop landing on a paved runway. Although the touchdown was uneventful, the airplane encountered a 10-knot crosswind during the landing roll, leading to a loss of directional control. The pilot reported no mechanical anomalies or malfunctions affecting the directional control system. This was labeled as *Loss of Directional Control (Cluster 0)*, but the model misclassified it as *Fuel-Related Issues (Cluster 2)* under Routine Violations. The fourth instance describes an accident involving an airplane landing at slow speed, that

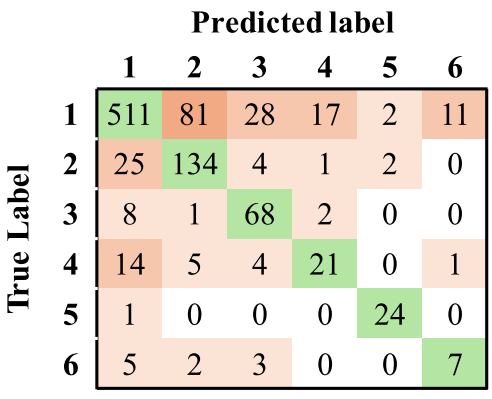


Fig. 7. Confusion matrix of test predictions.

Table 7
Examples of errors made by the model.

Instance		True	Predicted
1	**Label**	**5**	**3**
	accident description: reported lowered landing gear 5 miles airport flying instrument approach. short final turning airplane's positioning lights landing gear indicator lights would brighten, propellers struck runway, indicating likely loss situational awareness airplane's altitude runway. airplane skidded belly runway. examination airplane revealed wing spar sustained substantial damage. nose landing gear (nlg) doors sustained scraping damage consistent nlg retracted upon touchdown. postaccident examinations, main landing gear nlg lowered raised normally, landing gear indicator lights operated normally. landing checklist accomplished landing gear lowered airplane landing configuration. short final, doubts landing gear configuration, go-around accomplished. given ensured landing gear down, conducted go-around..		
2	**Label**	**0**	**3**
	accident description: reported landing flare experienced believed wind shear dust devil. airplane touched lifted 20 40 feet wind suddenly shifted tailwind airplane stalled. stated right wing dropped airplane entered approximate 45 60 degree angle bank. corrected bank angle leveled wings failed apply full power completely recover stall. airplane subsequently impacted runway nose down, bounced, settled runway. firewall fuselage substantially damaged. reported preimpact mechanical malfunction failures airplane would precluded normal operation.		
3	**Label**	**0**	**2**
	accident description: pilot, flying tailwheel-equipped airplane, performing full-stop landing paved runway. although touchdown uneventful, reported that, landing roll, lost control airplane encountered crosswind 10 knots. lost control, airplane ground looped, left wing struck runway sufficient force bend wing spars. reported mechanical anomalies malfunctions airplane's directional control system.		
4	**Label**	**4**	**1**
	accident description: tailwheel-equipped airplane reported that, landing grass airstrip, landing roll slow speed, airplane veered left ran runway. right-wing substantially damaged collided tree. reported preaccident mechanical malfunctions failures airplane would precluded normal operation.		

veered to the left and ran off the runway, ultimately colliding with a tree, causing substantial damage to the right wing. The pilot reported no preaccident mechanical malfunctions or failures that would have impacted the airplane's operation. This incident was labeled as a *Taxiing Collision (Cluster 4)* under *Perceptual Errors*, but the model misclassified it as *Hard Landings (Cluster 1)*. The misclassification likely stems from the model associating the damage to the right wing and the off-runway excursion with a hard landing, even though the description explicitly mentions a slow-speed landing roll on a grass airstrip and a tree.

While HFACS is a valuable framework for classifying human errors, it relies on manual categorization and does not inherently analyze large-scale accident narratives. Our methodology complements HFACS by enabling scalable, data-driven identification of pilot failure patterns. Future research could integrate HFACS categories into machine learning models to enhance human factor analysis further.

These results highlight the potential of NLP models in systematically identifying pilot failures in aviation accidents, even in the presence of overlapping and nuanced descriptions. Despite challenges in differentiating certain categories, the model demonstrated strong performance in well-defined failure types and provided valuable insights into patterns of misclassification. Notably, the framework offers a scalable and data-driven approach that supports expert analysis, enabling more consistent and objective classification of pilot errors. Furthermore, the adaptability of the proposed approach suggests that it could be fine-tuned for specific aircraft types or regulatory contexts, enhancing its applicability in targeted safety assessments. Future improvements, such as incorporating additional contextual features and refining preprocessing methods, could further enhance classification accuracy. Ultimately, this study underscores the value of integrating NLP with traditional investigative methods to improve the understanding of human factors in aviation safety and inform more effective risk mitigation strategies.

It is important to note that while our study analyses aviation accidents to identify key contributing pilot errors, the same framework could be applied to investigate accidents within specific aircraft categories, considering their unique regulatory and operational environments. Different aircraft types (e.g., commercial vs. private) are subject to distinct oversight structures that influence accident investigations, training standards, and safety protocols. For instance, commercial pilots undergo more rigorous training and fatigue management measures, which impact how human factors contribute to accidents [7,62]. Moreover, operational differences, such as single-pilot operations in general aviation versus multi-crew coordination in commercial aviation, affect pilot workload and error mitigation strategies [63]. The level of automation and aircraft complexity further modulates human error risks, with commercial aircraft relying heavily on automated systems, whereas private aircraft require more manual control, increasing cognitive demands during high-workload situations [64].

4.3. Comparison with the structural topic model (STM)

The coherence results (c_v) obtained for multiple STM configurations are summarized in Table 8, varying the minimum term frequency threshold (lower.thresh) and number of topics (K). Across all configurations, STM exhibited coherence values in the range 0.415–0.444, consistently lower than those observed for topic modelling based on BERT embeddings (0.505), and LDA (0.454). The best-performing STM configuration was obtained for $K = 10$ and lower.thresh $= 5$, yielding $c_v = 0.444$.

Moreover, the similarity matrix (Table 9) shows that the topic partitions produced by BERT-based and STM topic modelling exhibit only limited alignment. With the exception of one topic pair (BERT topic 2 ↔ STM topic 2, cosine $= 0.402$), all other similarities remain below 0.20, and most remain close to zero. This indicates that STM and BERT-based topic modelling capture distinct thematic structures: STM emphasizes lexical overlap, while BERT captures semantic proximity across lexically diverse descriptions.

Overall, these results confirm that topic modelling based on BERT embeddings achieves comparable or higher semantic coherence while providing more interpretable, semantically cohesive clusters suited to labeling the subcategories of pilot failure.

5. Conclusion

We have presented a methodology to identify pilot failures contributing to aviation accidents. Human factors are among the primary causes of these accidents, and their underlying reasons are typically described in accident investigation reports. These reports provide valuable insights for decision-making and uncovering root causes, ultimately reducing the potential recurrence of such events. Our methodology involves training a state-of-the-art NLP model and applying topic modeling to its contextual embeddings in order to semi-automated the annotation of the accident dataset that is a critical step for developing an accurate and efficient classifier. Using the publicly available NTSB database, we successfully categorized the data into seven meaningful

Table 8
STM coherence results for different configurations of minimum term frequency (lower.thresh) and number of topics (K). The configuration with $K = 10$ and lower.thresh $= 5$ achieved the highest coherence (0.444).

lower thresh	*K*	c_v	n_{topics}
5	6	0.429	6
5	**10**	**0.444**	**10**
3	6	0.425	6
3	10	0.428	10
2	6	0.415	6
2	10	0.424	10

Table 9
Cosine similarity matrix between BER-based and STM topic-term vectors.

	STM	stm_0	stm_1	stm_2	stm_3	stm_4	stm_5
BERT							
$bert_0$		0.000	0.000	0.000	0.000	0.000	0.105
$bert_1$		0.000	0.000	0.000	0.095	0.000	0.000
$bert_2$		0.000	0.000	0.402	0.000	0.000	0.000
$bert_3$		0.000	0.183	0.000	0.000	0.000	0.000
$bert_4$		0.000	0.000	0.000	0.000	0.092	0.000
$bert_5$		0.000	0.000	0.000	0.000	0.000	0.000

clusters after preprocessing, enabling multiclassification labeling.

The performance of the classifier was satisfactory, and while there were occasional misclassifications resulting from similarities in descriptions across different clusters, the trained model holds significant value in identifying pilot failures contributing to accidents. These findings further reinforce the effectiveness and potential of the model in enhancing accident investigation and analysis processes. While direct validation with industry investigators is valuable, validation in data-driven research extends beyond expert agreement. Our methodology has been validated through multiple approaches, including internal performance metrics (precision, recall, and F1-score), alignment with structured frameworks such as HFACS, and empirical misclassification analysis. Additionally, the model's ability to uncover systematic patterns in pilot failures provides actionable insights for developing targeted safety measures by identifying pilot failures, such as skill-based errors, routine violations, and perceptual errors, reinforcing its practical relevance. Future studies may further validate this approach by applying it to investigator-reviewed datasets and extending it to multi-label classification to better capture the multifactorial nature of aviation accidents.

The trained model can be useful because these descriptions are usually quite long; thus, the automatic identification of the pilot failures that led to the accident may spare experts' time in accident analysis. In addition, the model serves as a valuable decision-support tool by providing a preliminary assessment of potential causes even when accident descriptions are incomplete. These contributions hold practical significance, enabling experts to not only identify common human failure patterns but also propose precise preventive measures and effective training strategies to mitigate the risk of human error.

This study focuses on pilot failures as a representative category of human error to enable consistent supervised training. This choice does not imply that accidents are mono-causal; rather, it provides a stable representation for model development. The full posterior topic distribution is preserved and can be used to identify secondary causes, quantify topic co-occurrence, and flag ambiguous narratives for expert review. Future work will extend this approach to multi-label classification and hierarchical HFACS assignment, enabling more explicit modeling of multi-causal failure mechanisms.

The methodology is straightforward to apply and can be periodically used to identify emerging pilot failures and retrain the classifier. This ensures the continued relevance and effectiveness of the model in adapting to new trends and challenges in aviation safety, further supporting ongoing efforts to enhance accident prevention and operational safety.

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CRediT authorship contribution statement

July B. Macedo: Writing – review & editing, Writing – original draft, Software, Methodology. **Plínio M.S. Ramos:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Caio B.S. Maior:** Writing – review & editing, Writing – original draft, Conceptualization. **Márcio J.C. Moura:** Writing – review & editing, Validation. **Isis D. Lins:** Writing – review & editing, Writing – original draft, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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